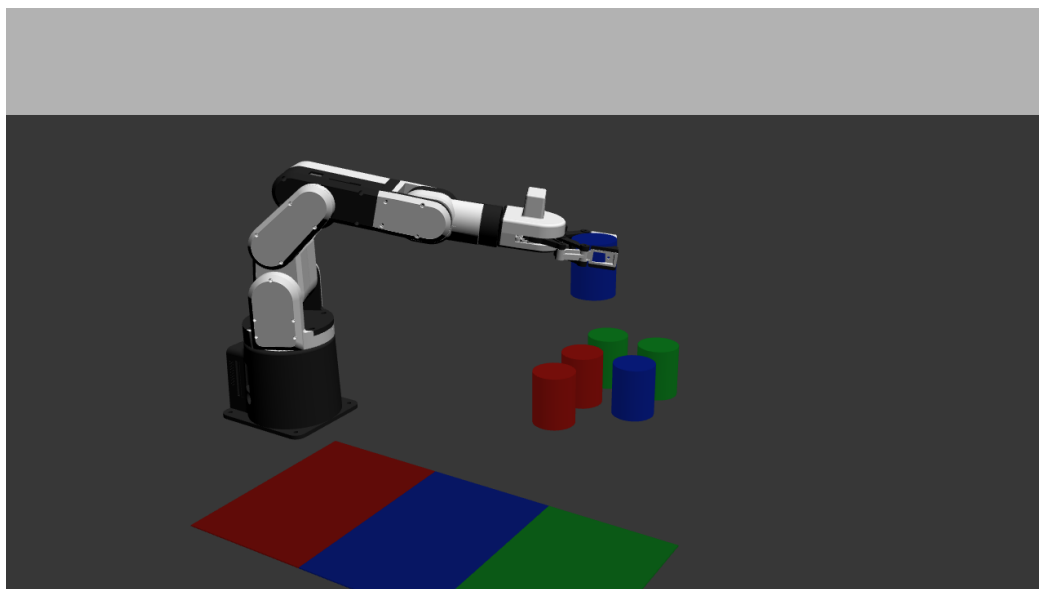


EXPERIMENT 3

Desktop Object Automatic Sorting and Organization

Experiment 3 · laboratory report

Group laboratory report



Course	Robotics Integration Group Project I
Group	Group 17: Yang Muqing, Lan Songyu, Zhang Kairui, and Zhao Hang
Report topic	Simulation development, sorting, and real-robot validation plan

ROS 2 HUMBLE · GAZEBO · mechARM 270

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GitHub: <https://github.com/9161942-boop/robotics-experiment-3>

Abstract

This experiment develops an automatic desktop object sorting system using ROS 2, Gazebo, and the mechArm 270. The system forms a complete perception–localization–decision–execution–acceptance loop: an overhead camera publishes the desktop image, a detection node outputs class labels, confidence scores, and bounding boxes, and the task node maps box centers to a fixed picking grid and generates an execution queue with target-bin information. The robot then performs grasping, transfer, release, and result logging through a state machine.

The system first uses six red and blue cylinders for quantitative simulation acceptance; the archived object-level record is 6/6. The scene is then extended to red, blue, and green objects to verify the added class, sorting region, waypoint sequence, and automatic playback interface. This report is organized into simulation development and real-robot validation phases, with quantitative results, process evidence, and the fields required for later real-robot acceptance listed separately.

Conclusion: the simulation phase completed the standard image interface, fixed-grid localization, class-to-bin mapping, grasp-and-release state machine, exception branches, and object-level logging. The red–blue baseline reached 6/6, the three-class extension completed scene and automatic-sequence integration, and the real-robot phase now has an implementable deployment and acceptance plan.

Keywords: ROS 2; Gazebo; mechArm 270; object detection; grid localization; sorting grasp; state machine; experiment acceptance

1 Experiment objectives

1.1 Task objective

Experiment 3 adds object detection and class decisions to fixed-point grasping. From the overhead image, the system obtains each object's class, confidence, and bounding-box center, determines its grid cell, calls the corresponding fixed picking point, and places the object in the target sorting region. The system processes multiple objects in a loop until the task queue is complete or no executable target remains.

Task control uses a feedback-enabled long-running interface and a state machine. Detection, grid assignment, grasping, placement, and exception logging are performed by software, while object-level states are saved during execution so that results can be reviewed from events and final poses.

1.2 Acceptance requirements and Group 17 implementation

Acceptance requirement	Implementation	Available evidence	Status
At least two object classes	Red–blue quantitative baseline; red–blue–green extension	Scene files, detection messages, sorting bins	Pass
At least six objects tested; at least five sorted correctly	Object-level grasp, release, and drop-point checks	Red–blue execution CSV: 6/6	Pass
Automatically select grid and sorting region after start	Map bbox center to grid_id; query target_bin by class	sorting_task and planning logs	Pass
Handle empty grids, missed detections, unreachable targets, and grasp failures	Filtering, skip, failure, and safe-return branches	skipped, failed, and ERROR_LOG	Pass
One Launch starts the complete simulation	Start world, camera, controllers, and task node together	sorting_gazebo.launch.py	Pass
Complete real-robot acceptance with six objects in at least two classes	Reuse the task interface; replace camera and execution layers	Real-robot deployment and acceptance checklist	Pending

2 Principle

2.1 Overall system architecture

The system consists of a perception layer, a task-decision layer, and a robot-execution layer. The perception layer first acquires an image; the visual detector outputs a class, bounding box, and confidence, which are then packaged as `Detection2DArray`. The decision layer performs confidence filtering, grid localization, class mapping, queue ordering, and exception branching. The execution layer controls the arm and gripper to approach, close, lift, transfer, release, and report results. The acceptance layer reads the GUI, stage states, event logs, final poses, and object-level CSV to form traceable evidence.



Fig. 1 Three-layer system architecture and object-level result loop

2.2 Visual Detection and Message Interface

The visual detector outputs a class, bounding-box center, and confidence, which enter the task node through `Detection2DArray`. The simulation monitor generates deterministic detections under the same contract, so the grid and task-control logic can be validated in Gazebo before the real camera and trained detector are connected to the same interface.

2.3 Key messages and interfaces

Interface or node	Type / form	Main responsibility
/sorting_camera/overhead/image_raw	sensor_msgs/Image	Simulation overhead RGB image
/sorting_scene/detections	vision_msgs/Detection2DArray	Class, confidence, bounding box, and object ID
/sorting_task/state	std_msgs/String	Stage feedback for scanning, filtering, planning, etc.
/sorting_task/result	std_msgs/String(JSON)	Locked object-level execution plan
/sorting_task/execution_state	std_msgs/String	Stage feedback for the long grasp task
/sorting_task/execution_result	std_msgs/String(JSON)	Final state, success count, and exception codes

Interface or node	Type / form	Main responsibility
/gazebo/set_entity_state	gazebo_msgs/SetEntityState	Reset, attachment-follow updates, and release

2.4 Exception-handling principle

The task node classifies exceptions as perception, localization, execution, or acceptance exceptions. An invalid object is excluded from the subsequent grasp queue, its reason is recorded, and other targets continue to be processed. If a safety pose, controller, or release-pose problem is detected, the executor first returns to a safe pose and then continues or stops according to the configuration.

Input condition	Program decision	Handling	Recorded field
No detection in the corresponding grid	empty grid	Do not create a task item	plan_count
score < 0.80	low confidence	Skip target	detect/skipped
Class not in configuration	unknown class	Skip target	unknown class
Nearest-grid distance > 48 px	outside grid	Skip target	outside grid
grid_id already occupied	duplicate grid	Keep the first valid target	duplicate grid
Release XY error > 0.07 m or z > 0.20 m	release failed	Record failure and move to the safe pose	exception/failed

3 Experimental environment and equipment

3.1 Simulation environment

Item	Configuration
Operating system and middleware	Ubuntu 22.04; ROS 2 Humble; Gazebo Classic 11
Robot model	mechArm 270 six-DOF arm, gripper, and ros2_control controller
Desktop and targets	0.72 m × 0.52 m desktop; cylinders with radius 0.022 m, height 0.055 m, and mass 0.04 kg
Picking grid	3 rows × 2 columns, six cells; x=0.24/0.30 m, y=0.12/0.18/0.24 m
Sorting bins	Red (0.14, -0.12), blue (0.26, -0.12), green (0.38, -0.12) m
Camera and thresholds	640×480 fixed overhead view; confidence ≥ 0.80; nearest-grid distance ≤ 48 px
Motion organization	manual_animation.json stores 27 waypoints; automatic playback has 34 steps; 2.0 s per step

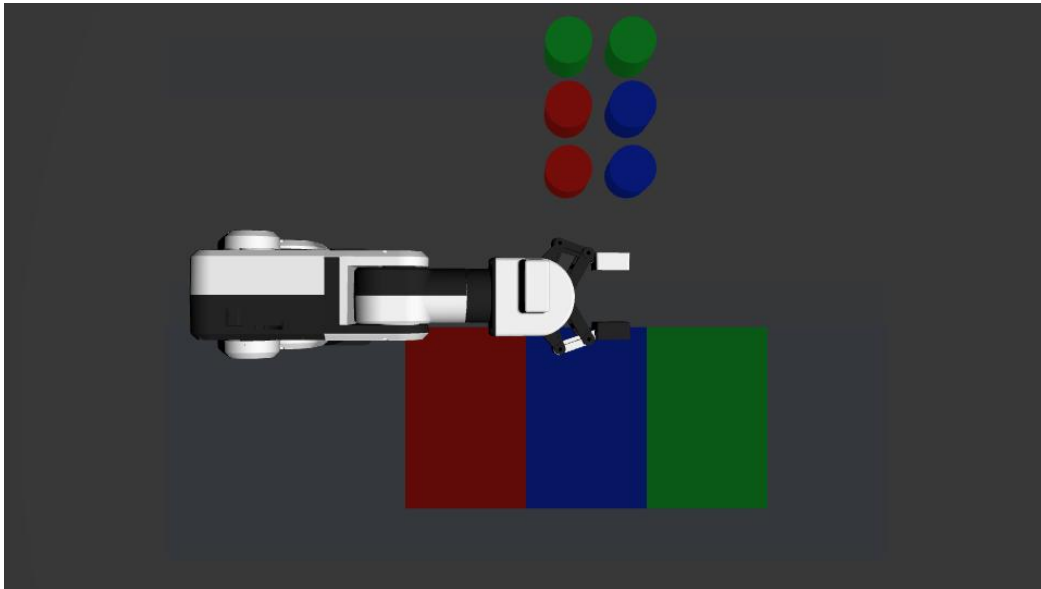


Fig. 2 Gazebo three-class scene: six picking cells and aligned sorting bins

3.2 Real-robot hardware and deployment configuration

The real-robot phase reuses the simulation task interface while replacing the camera driver, coordinate calibration, and robot-execution layer. The configuration below is prepared for deployment and acceptance; poses and thresholds must be recalibrated on the actual desktop before running the hardware.

Item	Real-robot implementation
Compute platform	Jetson edge platform running ROS 2, perception, and task-control nodes
Vision device	Fixed overhead RGB camera covering the six picking cells and sorting regions
Robot control	mechArm 270; pymycobot; serial port /dev/ttyAMA0; baud rate 1000000
Pose system	HOME safe pose, PICK1–PICK6 picking poses, and color-specific PLACE poses
Gripper parameters	Grasp opening, closure opening, and release wait time calibrated in single-target tests
Detection node	Connect to the preceding YOLOv8 class model and output class, confidence, and bounding box
Performance baseline	Reference record: Precision 0.658, Recall 0.748, mAP@0.5 0.766, about 15 FPS; re-record on hardware

4 Experimental procedure

4.1 Phase 1: simulation development

4.1.1 Visual Detection Model Training and External-Camera Validation

The perception module uses a YOLOv8n detector for three classes: eraser, lock, and stapler. The model outputs each class label, bounding box, and confidence. All images were collected by the group, with no public-dataset images mixed into the training set. The dataset includes single objects, mixed objects, different poses, positions, and lighting conditions; labellmg was used for manual annotation.

Training used a 640-pixel input, 60 epochs, batch size 16, and an RTX 4070 Laptop GPU. Mosaic composition, hue perturbation, random scaling, and random erasing were used during training, while the independently split validation images were kept unaugmented. The final weight is 6.25 MB (about 6.3 MB) with approximately three million parameters. The supplied training record reports $mAP50=0.995$ and $mAP50-95=0.995$ for the overall set and all three classes, with $precision=1.0$ and $recall=1.0$.

The iteration process addressed lock–stapler confusion, stapler–eraser confusion at specific angles, dilution of early samples after new data were added, and the gap between validation images and external-camera scenes. The corresponding actions were class-balanced sampling, same-frame comparison samples, a stronger classification-loss contribution, higher sampling weight for early samples, and targeted camera-data collection. Overall $mAP50-95$ improved from 0.938 to 0.995. In the supplied external-camera frame, all three classes are detected with confidence 0.89–0.94 and a displayed 31.2 FPS. The reported 1.1 ms is the single-model inference time, while 31.2 FPS is the end-to-end camera display rate.

For simulation, the monitor continues to generate deterministic `Detection2DArray` messages from Gazebo model states so that the task and execution interfaces remain repeatable. The trained weight is archived separately in `models/` for real-camera integration and hardware validation.

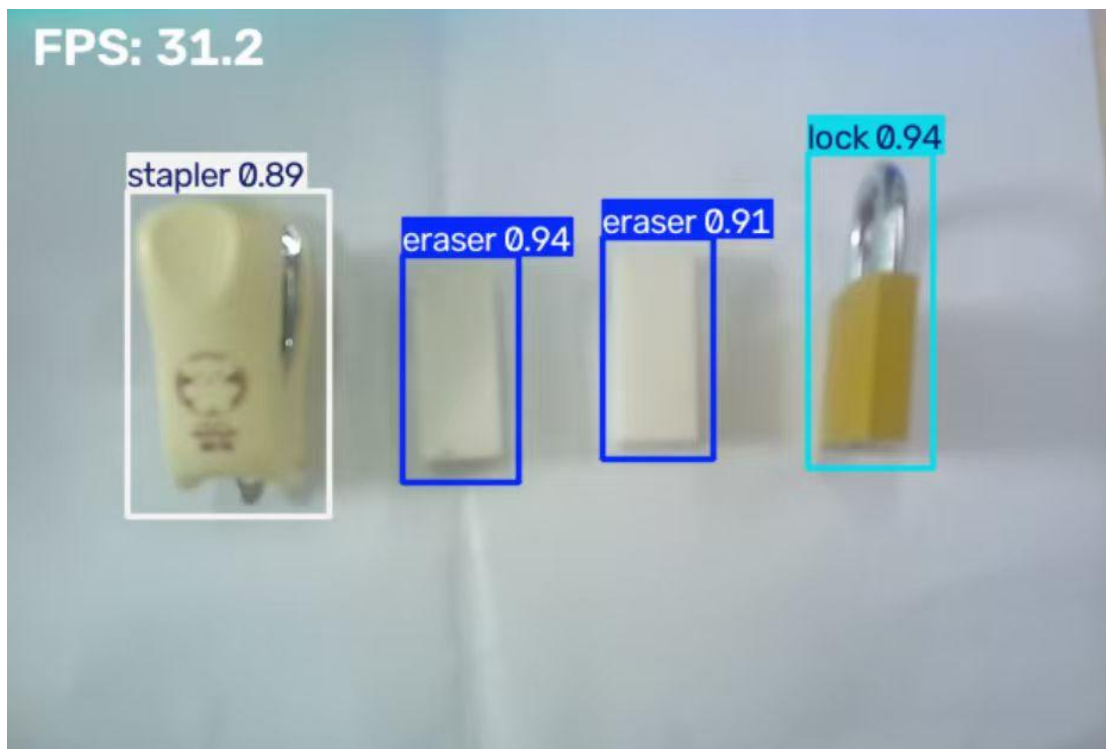


Fig. 3 External-camera detection of erasers, a lock, and a stapler

4.1.2 Scene construction and parameter synchronization

The Gazebo world loads the robot arm, gripper, desktop, a three-row/two-column picking grid, red–blue–green sorting bins, and an overhead camera. World coordinates, projected pixels, the grid array, and waypoint poses are calibrated as one set. The picking-cell centers correspond to pixel columns [212, 266, 320] and rows [186, 240, 294]. The projection is defined as $u=320+(y_w-0.18)\times 900$ and $v=240-(x_w-0.30)\times 900$.

4.1.3 Manual teaching and single-object debugging

Starting from initial or HOME, the team created the above, pick, lift, release_ready, release, and retreat poses in sequence. Only one group of joint or gripper parameters was changed at a time, and the actual relative position of the fingers and cylinder was checked in the GUI. After a single-point test confirmed collision-free motion, gripper closure, and target lifting, the pose was saved to `manual_animation.json`.



Fig. 4 Manual control panel, six-joint adjustment, and semantic waypoint editing

4.1.4 Automatic playback and one-command launch

auto_sequence_player reads the saved waypoints, recognizes the pick and release semantics, and plays the six target subsequences in a 34-step order. Before the formal run, the objects are reset, the controllers are checked, and only one Gazebo world instance is kept. A single Launch file starts the world, models, controllers, camera, and task nodes together.



Fig. 5 Six target subsequences and the 34-step automatic playback timing

4.1.5 Grasping, transfer, and release

The grasp sequence executes the safe pose, descent, closure, distance check, attachment, and lift in order. The transfer first passes through a safe intermediate pose and then reaches the target bin. During release, the current end-effector pose is read, X/Y alignment is maintained, the end effector moves briefly downward along Z, the attachment is removed, and the final model pose is read. Objects of the same class use different sub-drop points within the same bin.



Fig. 6 Gripper approach, closure, distance check, attachment, and lift



Fig. 7 Alignment above the sorting bin, vertical release, and drop-point acceptance

4.1.6 Simulation acceptance records

During execution, the system saves state, execution_state, execution_result, event logs, final poses, and an object-level CSV. The GUI view and result files are checked together to confirm that each target has left the desktop, entered the corresponding sorting region, and satisfied the drop-point condition.

4.2 Phase 2: real-robot validation

4.2.1 Hardware deployment and calibration

The perception, task, and execution nodes are deployed on Jetson. A fixed overhead RGB camera is installed, and a calibration board or homography establishes the pixel-to-desktop transformation. HOME, PICK1–PICK6, and the PLACE poses for each color are taught at low speed; approach height, gripper opening, release wait time, and emergency-stop distance are recorded.

4.2.2 Detection node and task interface

The YOLOv8 node outputs class labels, confidence scores, and bounding boxes, with message fields consistent with the simulation Detection2DArray. The task node continues to perform confidence filtering, grid mapping, class selection, and exception branching, while the executor replaces simulation control calls with mechArm hardware-control calls.

4.2.3 Validation sequence from one target to six targets

A single target is first used to verify detection, PICK, lift, and PLACE. A second class is then added, followed by the full six-target, two-class task. For each object, run_id, object_id, true class, predicted class, confidence, grid_id, target bin, grasp result, placement result, exception code, and timestamp are recorded. Empty grids, missed detections, grasp failures, and unreachable targets are tested separately.

4.2.4 Real-robot acceptance checklist

Check item	Pass condition	Record
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Check item	Pass condition	Record
Class and grid	Automatically detect at least two classes; unique mapping to six cells	Detection2DArray and task CSV
Grasp and placement	At least five of six targets enter the correct sorting region	Object-level execution table
Manual intervention	No manual class, grid, or bin selection after task start	Run video and logs
Exception handling	Empty grids and missed objects are skipped; failed targets return safely or stop	Exception codes and ERROR_LOG
Safety	No collision, joint-limit violation, or dangerous motion; emergency stop available	Safety checklist
Launch method	One Launch starts image, perception, robot, and task control	Launch output

5 Problem analysis and discussion

5.1 Simulation-stage problems

Observed phenomenon	Cause analysis	Fix and verification
Node exits after startup	Humble Python logger call did not match the current interface	Use an f-string; the node resumes continuous state publishing
Only two targets are planned	Old pixel-row coordinates remained after world-coordinate adjustment	Update to 186/240/294; assign a unique grid_id to all six targets
Gripper closes but target does not lift	The player sent only the gripper command and did not update the target relative pose	Add distance checks, attachment, and continuous following; the grasp_attached event appears
Objects scatter or same-class objects overlap	Unstable contact solving, fixed drop points, and lateral jumps	Stabilize model parameters, release vertically, and assign sub-drop points by class
Automatic playback fails after a repeated launch	The old Gazebo world and controllers still occupy resources	Keep one master only; replay the sequence in a clean instance

5.2 Real-robot-stage problems

Problem source	Possible symptom	Engineering treatment
Lighting and occlusion	Lower class confidence and jittering boxes	Fix exposure and add lighting; use confidence and multi-frame stability checks
Grid-mapping offset	Adjacent objects assigned to the wrong grid cell	Use a calibration board to compute a homography; record pixel error and coordinate version
Gripper grasp failure	Object slips or remains on the desktop after closure	Calibrate approach height, opening, closure speed, and lift wait per object
Serial or controller fault	Execution interruption or missing state feedback	Check connection state; return to HOME or trigger the emergency stop on failure
Simulation–hardware gap	Differences in trajectory, friction, mass, and latency	Start with a low-speed single target and expand gradually to all six targets

5.3 System limitations and improvement directions

The current grid scheme is based on a fixed desktop and pre-calibrated poses, which is suitable for this acceptance task. Arbitrary object locations would require continuous coordinate estimation and inverse-kinematics planning. Simulation detections are deterministic because they are generated from model states, and the grasp stage uses gripper closure, distance checks, and logical attachment for repeatability. Real-robot operation must remeasure visual error, gripper friction, approach height, release wait time, and safe speed.

Future versions can add multi-frame debouncing, run IDs, log rotation, object-occupancy states, dynamic-obstacle checks, and emergency-stop fallback while keeping the existing Detection2DArray, state-feedback, and object-level result fields unchanged.

6 Experimental results and acceptance evidence

6.1 Visual Detection and Interface Validation

On the independent original-image validation set, the overall result and all three classes reached $mAP50=0.995$ and $mAP50-95=0.995$, with precision=1.0 and recall=1.0. The external-camera frame in Fig. 3 shows all three classes with confidence values of 0.89–0.94 and a displayed rate of 31.2 FPS. This verifies that the detector output can enter the Detection2DArray interface; it is kept separate from the deterministic Gazebo sorting result and does not represent a real-robot grasping score.

6.2 Quantitative red–blue baseline with six objects

The object-level logs for the six red and blue objects contain the planned grid, target bin, grasp_attached, release, and object_complete events. The final state is passed with success_count=6. This satisfies the simulation acceptance requirement that at least five of six objects be sorted correctly.

Object	True / predicted class	Confidence	Grid	Target bin	Grasp	Place
object_blue_1	Blue / blue	0.99	r2c1	Blue	Pass	Pass
object_blue_2	Blue / blue	0.99	r3c2	Blue	Pass	Pass
object_blue_3	Blue / blue	0.99	r1c2	Blue	Pass	Pass
object_red_1	Red / red	0.99	r3c1	Red	Pass	Pass
object_red_2	Red / red	0.99	r1c1	Red	Pass	Pass
object_red_3	Red / red	0.99	r2c2	Red	Pass	Pass

6.3 Three-class extension

The extended scene adds two green cylinders and a green sorting bin. Three-class scene alignment, class mapping, waypoint naming, and the 34-step automatic playback sequence are completed. This phase shows that the added class uses the same task interface. Process evidence includes the overhead scene, control panel, waypoint list, and playback view; the object-level success rate is to be calculated from an independent three-class CSV.

6.4 Multi-layer evidence chain

The GUI view is used to inspect the geometric relationship between the gripper and target, stage states locate the current action, event logs confirm grasp and release triggers, final model poses check the bin drop point, and the object-level CSV summarizes the state and exception code of every target.



Fig. 8 GUI, state, event, pose, and object-level CSV form the acceptance evidence chain

7 Group responsibilities

The group divided the work into perception interfaces, task decisions, robot execution, and system integration. All members integrated their modules under one Launch file, one message-field convention, and one object-level result table.

Member	Main responsibilities	Deliverables
Yang Muqing	Gazebo scene, six-cell grid, and bin adjustment; manual teaching; grasp, attachment, vertical release, and evidence review; execution-side interface testing with shared detection results	sorting_desktop.world; manual_animation.json; GUI and CSV verification
Lan Songyu	Object-detection interface, class fields, and detection-result organization	Detection2DArray messages, class configuration, and detection-test records
Zhang Kairui	Grid mapping, task queue, state machine, and exception branches	sorting_task, state topics, and skipped/failed logs
Zhao Hang	ROS 2 Launch, execution interface, real-robot deployment preparation, and material consolidation	sorting_gazebo.launch.py, execution node, and real-robot acceptance checklist
Group collaboration	Three-class visual-detection data collection, annotation, YOLOv8n training, evaluation, and external-camera validation	models/best_yolov8n_eraser_lock_stapler.pt; models/README.md; detection screenshot

8 Conclusions and submitted materials

8.1 Conclusions

Experiment 3 extends the fixed-point transfer of Experiment 2 into a desktop organization task that can detect, localize, classify, execute, and record. The system integrates the ROS 2 image and detection interfaces, fixed-grid localization, class-to-bin mapping, the grasp-execution state machine, exception branches, and object-level logs. The six-object red–blue simulation baseline reaches 6/6, and the red–blue–green scene together with the 34-step automatic demonstration is complete.

During simulation, the group established the interface boundaries and acceptance fields needed for migration to the real robot. The hardware work will proceed in the order of camera calibration, low-speed single-target grasping, a

second class, the six-target closed loop, and exception-safety tests, with emphasis on detection stability, grid mapping, gripper opening, release points, controller state, and emergency-stop behavior.

8.2 Submission-material mapping

Material	Path or content
Complete ROS 2 project	src/mecharm_fixed_pick_sim/
Single Launch	launch/sorting_gazebo.launch.py
Scene and grid	worlds/sorting_desktop.world; config/sorting_scene.yaml
Task control	sorting_task.py; sorting_executor.py
Manual and automatic teaching	manual_control_panel.py; auto_sequence_player.py; manual_animation.json
Results and exception logs	~/ros/mecharm_sorting_task.csv; ~/ros/mecharm_sorting_execution.csv; ERROR_LOG.md
Report files	Experiment 3 group and personal report DOCX/PDF
Detection model and camera evidence	models/best_yolov8n_eraser_lock_stapler.pt; models/README.md; evidence/model_detection_realtime.png

Scope of results: the red–blue 6/6 result uses the archived object-level simulation record; the three-class content uses scene, waypoint, and automatic-playback evidence; the real-robot section provides deployment, calibration, execution, and acceptance fields.

8.3 GitHub synchronization record

The Experiment 3 project, evidence, and reports have been synchronized to the GitHub repository: <https://github.com/9161942-boop/robotics-experiment-3>.

The repository is organized into code, models, evidence, report, and videos directories, with a commit manifest, run instructions, and multiple commits retained. The two Gazebo demonstration videos are stored at `videos/experiment3_gazebo_sorting_run_01.webm` and `videos/experiment3_gazebo_sorting_run_02.webm`. The reports, PDFs, and screenshots are also included as GitHub evidence so that the submitted content, source, and commit times can be reviewed from the repository page.

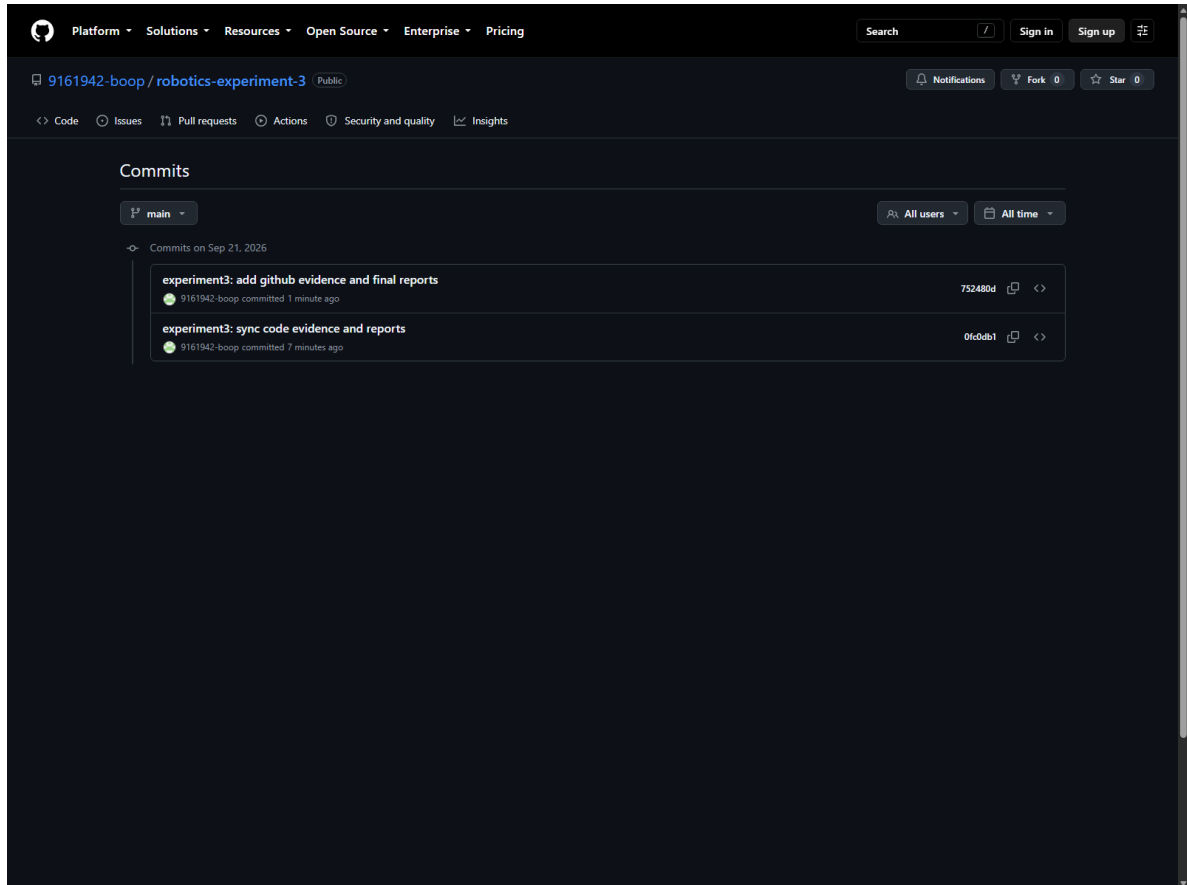


Fig. 9 GitHub Experiment 3 repository commit history